

Operating Systems and Computer Security

**CSCI 1101 Introduction to
Computer Science**

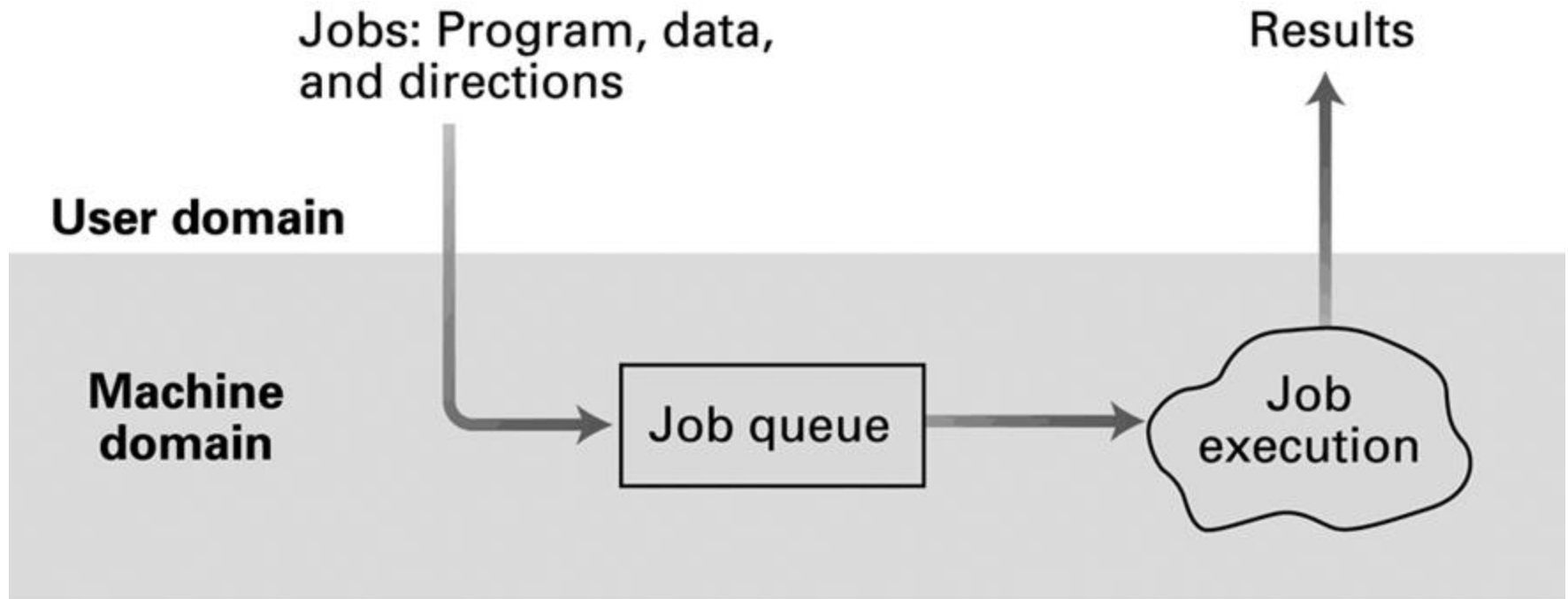
Functions of Operating Systems

- **Operating system (OS):** acts as the “master controller” of the computer system
 - Oversees operation of computer
 - Stores and retrieves files
 - Schedules programs for execution
 - Coordinates the execution of programs
 - Manages the system’s resources (CPU, memory, hardware)

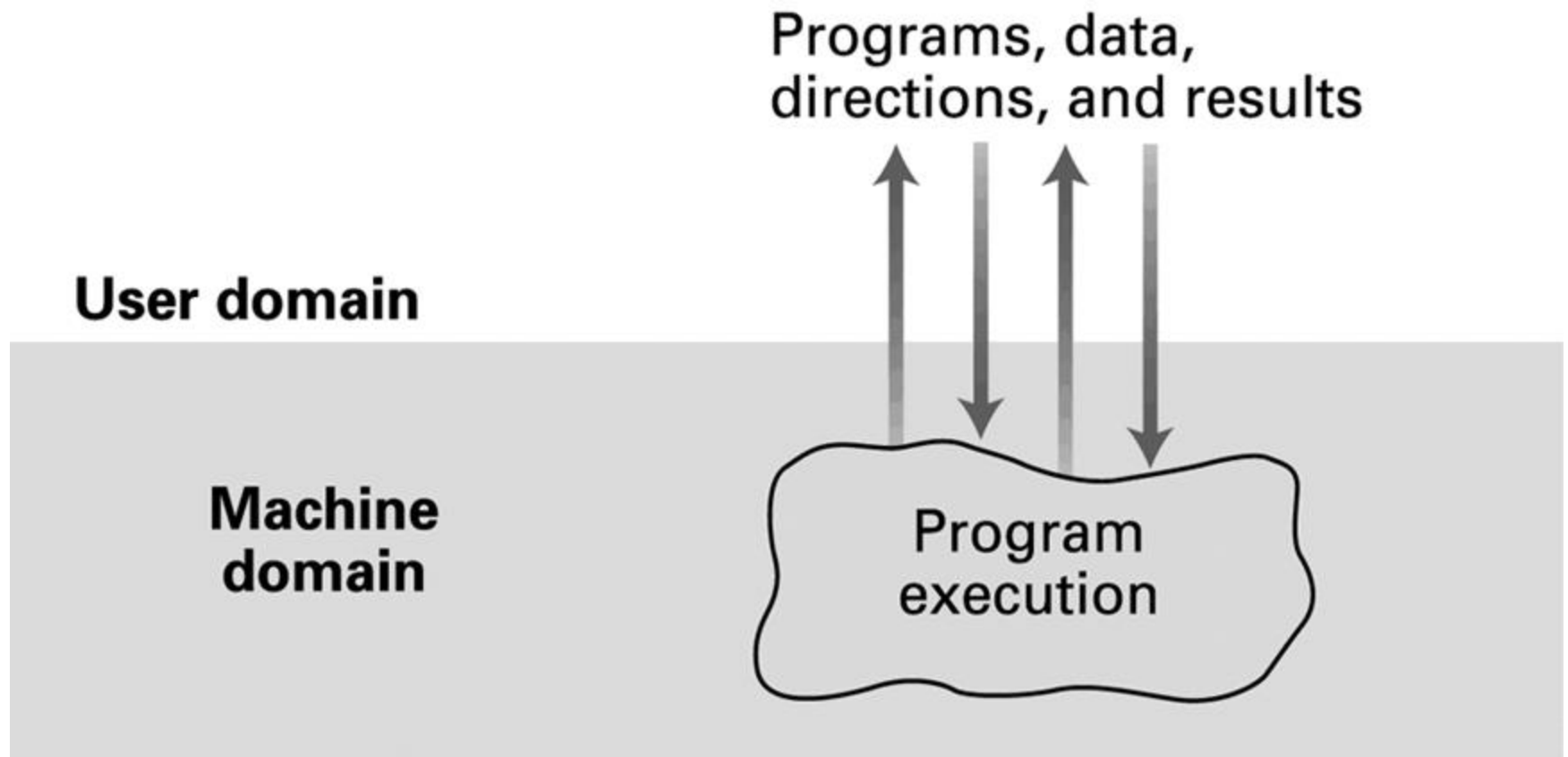
Evolution of Operating Systems

- Batch processing
- Interactive processing
 - Requires real-time processing
- Time-sharing/Multitasking
 - Implemented by Multiprogramming
- Multiprocessor machines

Batch Processing



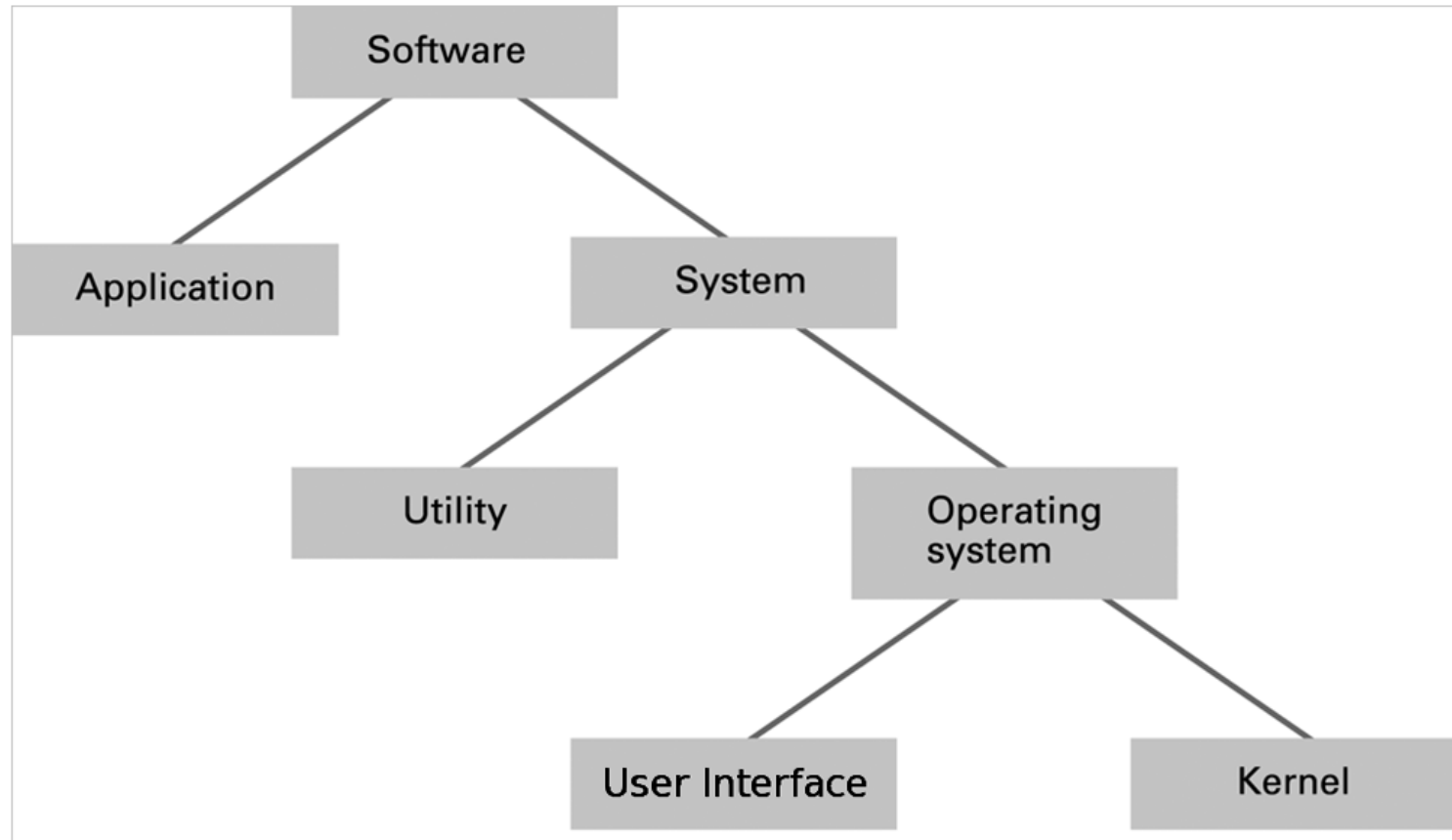
Interactive Processing



Types of Software

- **Application software**
 - Performs specific tasks for users
- **System software**
 - Provides infrastructure for application software
 - Consists of operating system and utility software

Software Classification

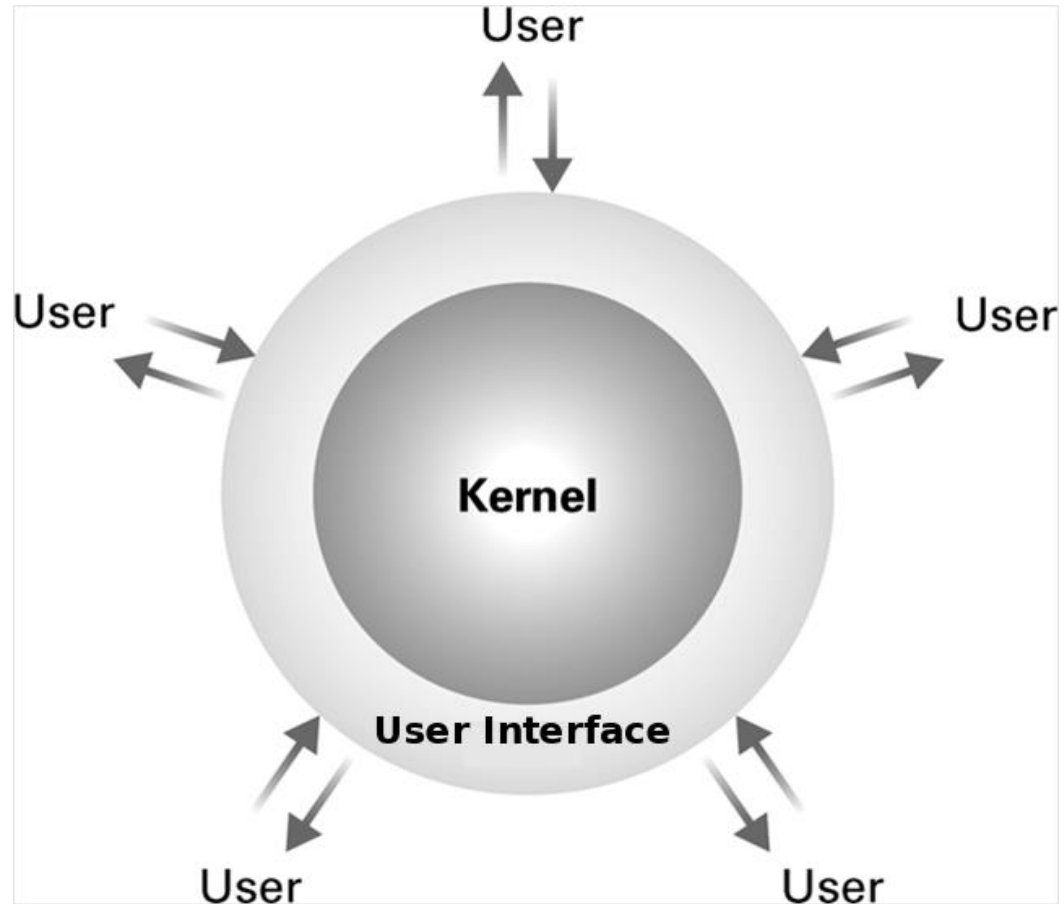


Operating System Components

- **User Interface (UI):** Communicates with users
 - Text based (Shell)
 - Graphical user interface (GUI)
- **Kernel:** Performs basic required functions
 - File manager
 - Device drivers
 - Memory manager
 - Scheduler and dispatcher

Role of the UI

- The user interface (UI) acts as an intermediary between users and the OS kernel



File Manager

- **Directory (or Folder):** A user-created bundle of files and other directories (subdirectories)
- **Directory Path:** A sequence of directories within directories

File Types

- **File Type:** The kind of information contained in a document
- **File format:** The underlying encoding of the file
- **File extension:** Indicates what software was used to create the file; indicates file format and implies file type

File Types

Extensions	File type
txt	text data file
mp3, au, wav	audio file
gif, tiff, jpg	image file
doc, wp3	word processing document
java, c, cpp	program source files

Some common file types and their extensions

What kind of files are

- *Chapter.doc*
- *Figure1.jpg*
- *Interview.wav*
- *MyFavorite.mp3*

What's the advantage of using the appropriate extension?

Protecting Files

File protection: The process of limiting file access

- In multiuser systems, file protection is of primary importance
- We don't want one user to be able to access another user's files unless the access is specifically allowed
- A file protection mechanism determines who can use a file and for what general purpose

Why is file protection important?

File Protection

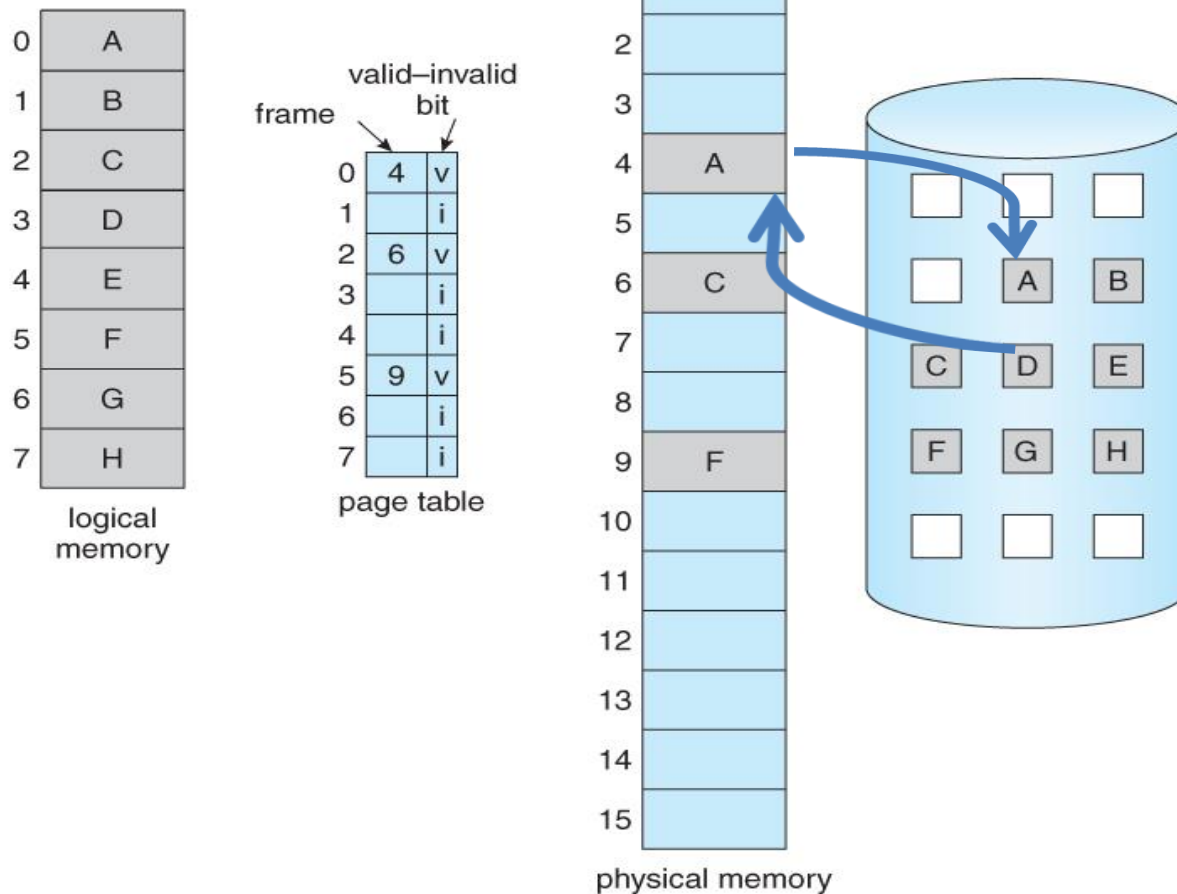
An example of a file protection scheme is the file settings in the Unix operating system, which are divided into three categories

	Read	Write/Delete	Execute
Owner	Yes	Yes	Yes
Group	Yes	Yes	No
World	Yes	No	No

Memory Manager

- Allocates space in main memory
- May create the illusion that the machine has more memory than it actually does (**virtual memory**) by playing a “shell game” in which blocks of data (**pages**) are shifted back and forth between main memory and mass storage (page swapping)
- **Thrashing**: Inefficient processing caused by constant page swaps

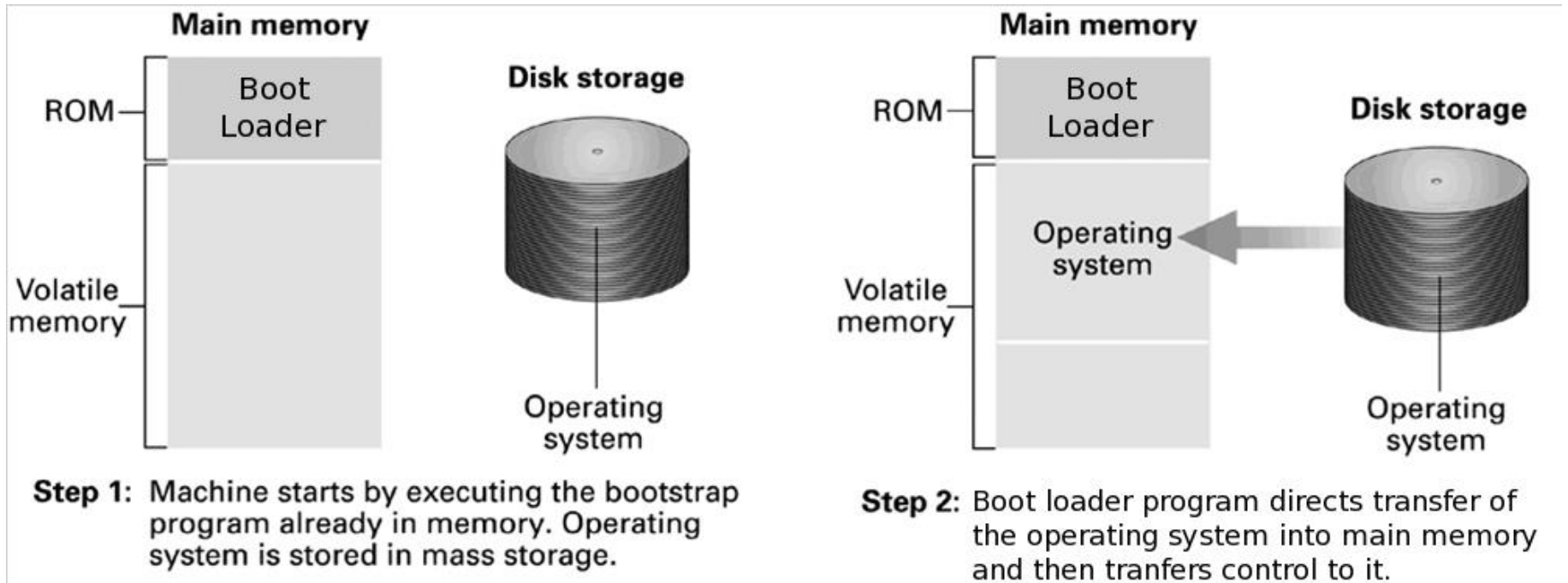
Virtual Memory (page swapping)



Getting it Started (Bootstrapping)

- **Boot loader:** Program in ROM (example of firmware)
 - Run by the CPU when power is turned on
 - Transfers operating system from mass storage to main memory
 - Executes jump to operating system

The Booting Process

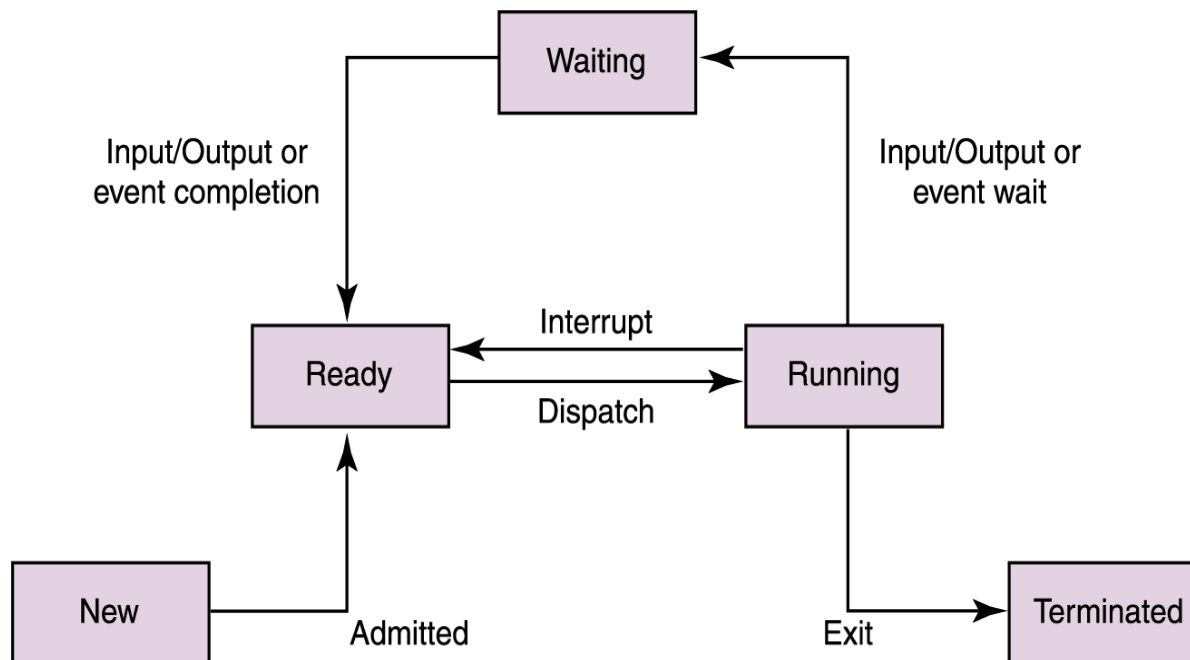


Processes

- **Process:** The activity of executing a program
- **Process State:** Current status of the activity
 - Program counter
 - General purpose registers
 - Related portion of main memory

Process States

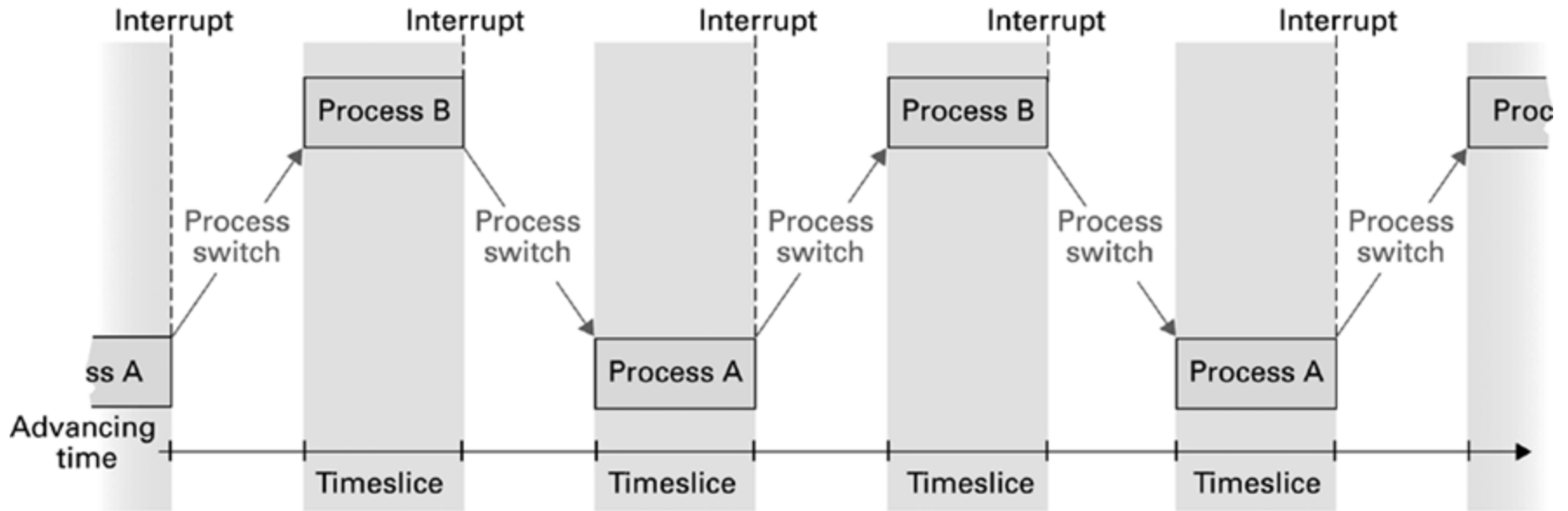
During its “life cycle,” a process is in one of the following states:



Process Administration

- **Scheduler:** Adds new processes to the process table and removes completed processes from the process table
- **Dispatcher:** Controls the allocation of time slices to the processes in the process table
 - The end of a time slice is signaled by an interrupt.
 - **Context (or process) switch:** saving the state of the process that is being interrupted and loading the state of the process that will execute in the next time slice

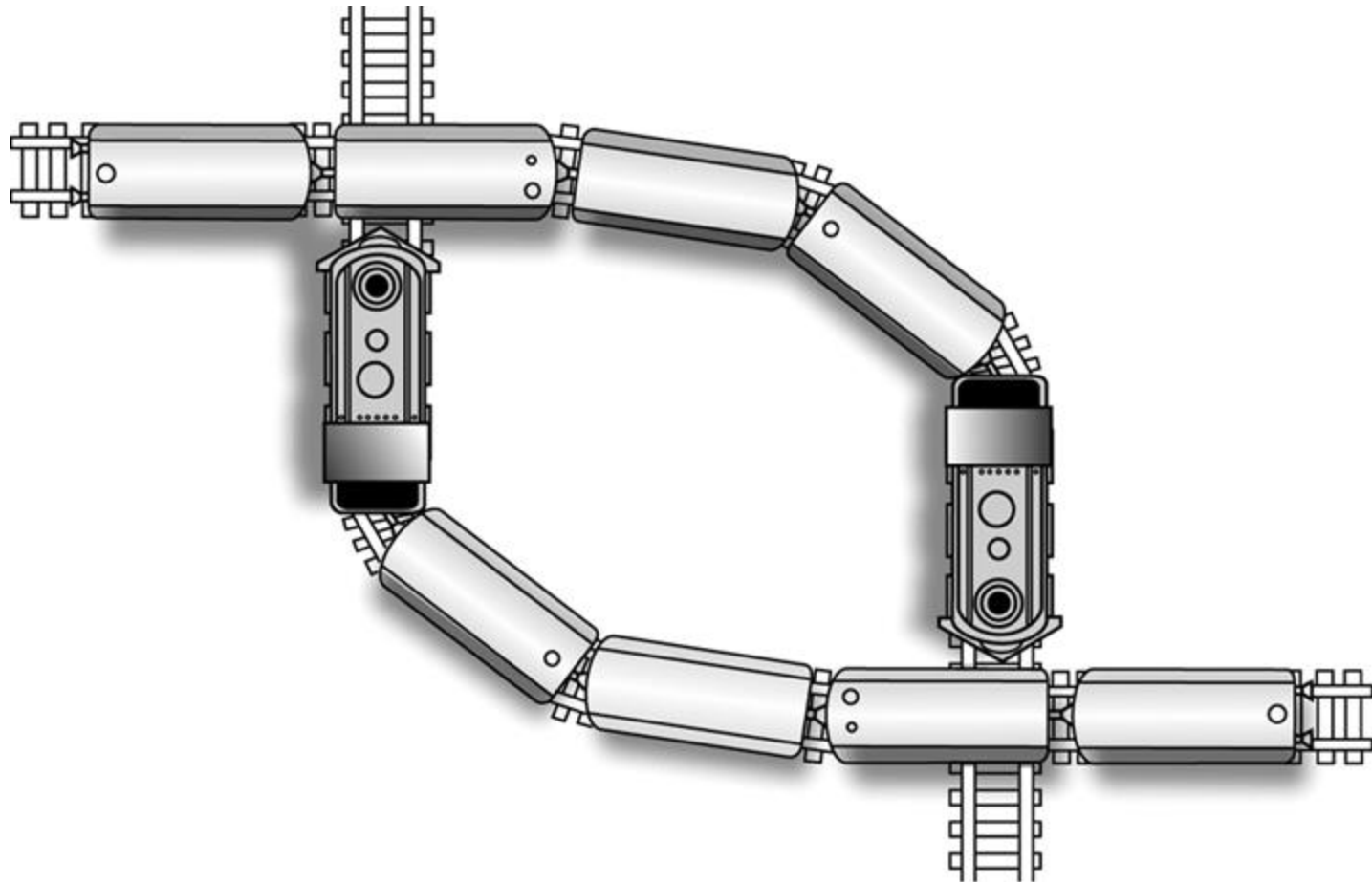
Time-sharing Between Process A and Process B



Deadlock

- Processes block each other from continuing
- Conditions required for deadlock
 1. Competition for non-shareable resources
 2. Resources requested on a partial basis
 3. An allocated resource cannot be forcibly retrieved

A deadlock resulting from competition for nonshareable railroad intersections



Computer Security

- Security involves protecting the computer from threats from
 - Within the system
 - Outside the system

Computer Security

- Attacks from within
 - Problem: Unruly processes
 - Counter measures: Control process activities via privileged modes and privileged instructions

Computer Security

- Attacks from outside
 - Problems
 - Insecure passwords
 - Sniffing software
 - Malicious code
 - Counter measures
 - Auditing software
 - Firewalls

Computer Security

- **Authentication credentials**
 - Information users provide to identify themselves for computer access
 - Methods
 - **User knowledge** name, password, PIN
 - **Smart card** card with embedded memory chip used for identification
 - **Biometrics** human characteristics such as fingerprints, retina or voice patterns

Computer Security

■ Malicious code

- A computer program that attempts to bypass appropriate authorization and/or perform unauthorized functions
- Types of malicious programs
 - **Worm:** stands alone, targets network resources
 - **Trojan horse:** disguised as benevolent resource
 - **Virus:** self-replicating
 - **Logic bomb:** set up to execute at system event
 - **Blended threats :** combine attributes of several types

Computer Security

■ Security attacks

- An attack on the computer system itself
- Types of attacks
 - **Password guessing** obvious
 - **Phishing** trick users into revealing security information
 - **Spoofing** malicious user masquerades as authorized user
 - **Back door** unauthorized access to anyone who knows it exists

Computer Security

- **Network-based security attacks**
 - **Denial-of-service** attack that prevents authorized user from accessing the system (typically directed at web sites)
 - **Man-in-the-middle** network communication is intercepted in an attempt to obtain key data